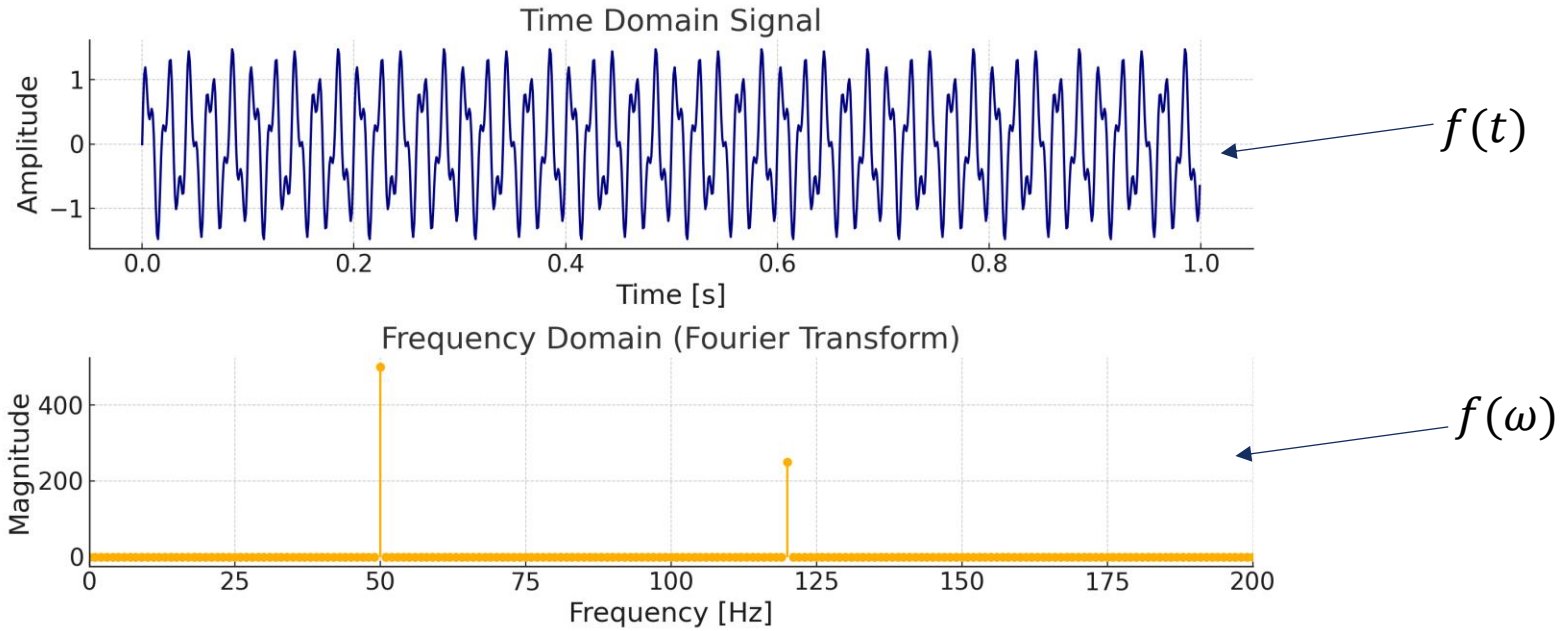
Fourier Transform

DR. E. M. D. SIRIWARDANE
DEPARTMENT OF PHYSICS
UNIVERSITY OF COLOMBO

Fourier Transform

The Fourier Transform is a mathematical tool used to analyze and represent functions or signals in terms of their frequency components.

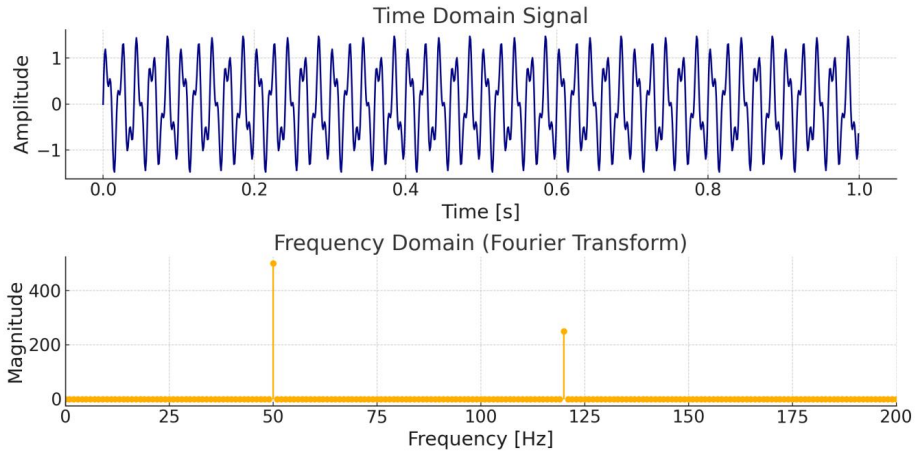
It transforms a time-domain signal into its frequency-domain representation.



Fourier Transform

For a continuous-time signal $f(t)$, the Fourier Transform $F(\omega)$ is defined as:

$$\boxed{F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt}$$



$f(t)$: original signal (time domain)

$F(\omega)$: transformed signal (frequency domain)

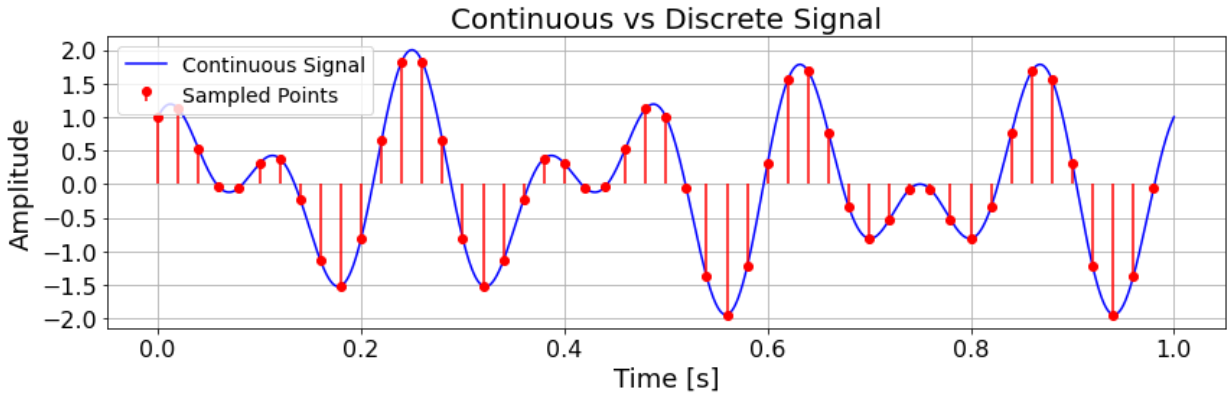
ω: angular frequency

i: imaginary unit $\quad i = \sqrt{-1}$

The **inverse Fourier Transform** recovers the original signal:

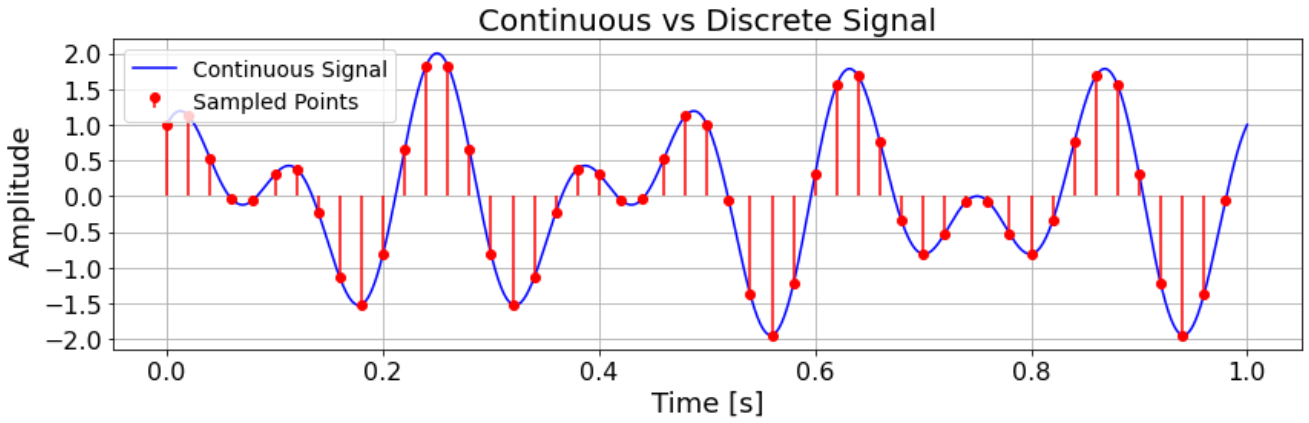
$$\boxed{f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} dt}$$

Discrete Fourier Transform (DFT)



- In practice, signals are recorded as discrete samples (e.g., using sensors, microphones, or ADCs).
- Computers operate on **finite, digital data**.
- The DFT is designed to analyze a finite sequence of numbers, making it suitable for computational use.
- It allows numerical frequency analysis on actual data stored in memory.

Discrete Fourier Transform (DFT)



The continuous Fourier transform is given by

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

The discrete Fourier transform is given by

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i \frac{kn}{N}}$$

The above expression is evaluated for N number of samples

$\frac{k}{N}$ is corresponding to frequency

n is corresponding to time t

Discrete Fourier Transform (DFT)

$$\boxed{X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i \frac{kn}{N}}}$$

$$b_n = \frac{2\pi kn}{N} \rightarrow X_k = \sum_{n=0}^{N-1} x_n e^{-ib_n}$$

Euler's formula

$$e^{i\theta} = \cos\theta + i\sin(\theta) \qquad i = \sqrt{-1}$$

Therefore, e^{-ib_n} can be written as

$$e^{-ib_n} = \cos(b_n) - i\sin(b_n)$$

$$X_k = \sum_{n=0}^{N-1} x_n(\cos(b_n) - i\sin(b_n))$$

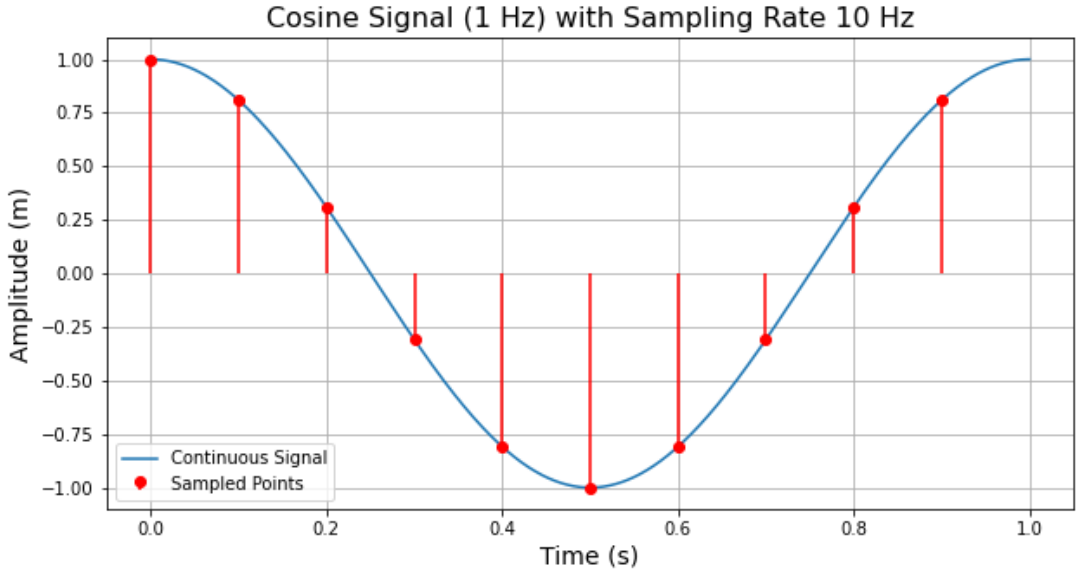
Therefore, we can write the above expression with sine and cosine components as follows

$$X_k = \sum_{n=0}^{N-1} x_n\left(\cos\left(\frac{2\pi kn}{N}\right) - i\sin\left(\frac{2\pi kn}{N}\right)\right)$$

$$X_k = \sum_{n=0}^{N-1} x_n(A - iB)$$

Complex numbers

Example 1



Sample 0: Time = 0.00 s, Value = 1.000
Sample 1: Time = 0.10 s, Value = 0.809
Sample 2: Time = 0.20 s, Value = 0.309
Sample 3: Time = 0.30 s, Value = -0.309
Sample 4: Time = 0.40 s, Value = -0.809
Sample 5: Time = 0.50 s, Value = -1.000
Sample 6: Time = 0.60 s, Value = -0.809
Sample 7: Time = 0.70 s, Value = -0.309
Sample 8: Time = 0.80 s, Value = 0.309
Sample 9: Time = 0.90 s, Value = 0.809

$$X_k = \sum_{n=0}^{N-1} x_n \left(\cos\left(\frac{2\pi kn}{N}\right) - i \sin\left(\frac{2\pi kn}{N}\right) \right)$$

When $k = 0$

$$X_0$$

$$= 1.00 \left(\cos\left(\frac{2\pi \times 0 \times 0}{10}\right) - i \sin\left(\frac{2\pi \times 0 \times 0}{10}\right) \right)$$

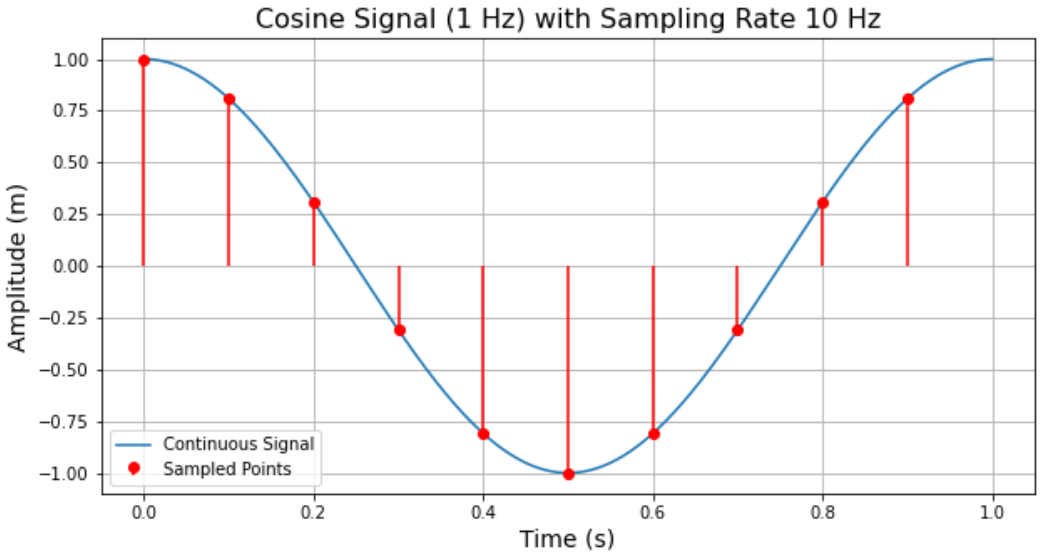
$$+ 0.809 \left(\cos\left(\frac{2\pi \times 0 \times 1}{10}\right) - i \sin\left(\frac{2\pi \times 0 \times 1}{10}\right) \right)$$

$$+ 0.309 \left(\cos\left(\frac{2\pi \times 0 \times 2}{10}\right) - i \sin\left(\frac{2\pi \times 0 \times 2}{10}\right) \right)$$

$$+ \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

$$+ 0.809 \left(\cos\left(\frac{2\pi \times 0 \times 9}{10}\right) - i \sin\left(\frac{2\pi \times 0 \times 9}{10}\right) \right)$$

Example 1



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Sample 9: Time = 0.90 s, Value = 0.809

$$X_k = \sum_{n=0}^{N-1} x_n \left(\cos\left(\frac{2\pi kn}{N}\right) - i \sin\left(\frac{2\pi kn}{N}\right)\right)$$

When $k = 0$

$$X_1$$

$$= 1.00\left(\cos\left(\frac{2\pi \times 1 \times 0}{10}\right) - i \sin\left(\frac{2\pi \times 1 \times 0}{10}\right)\right)$$

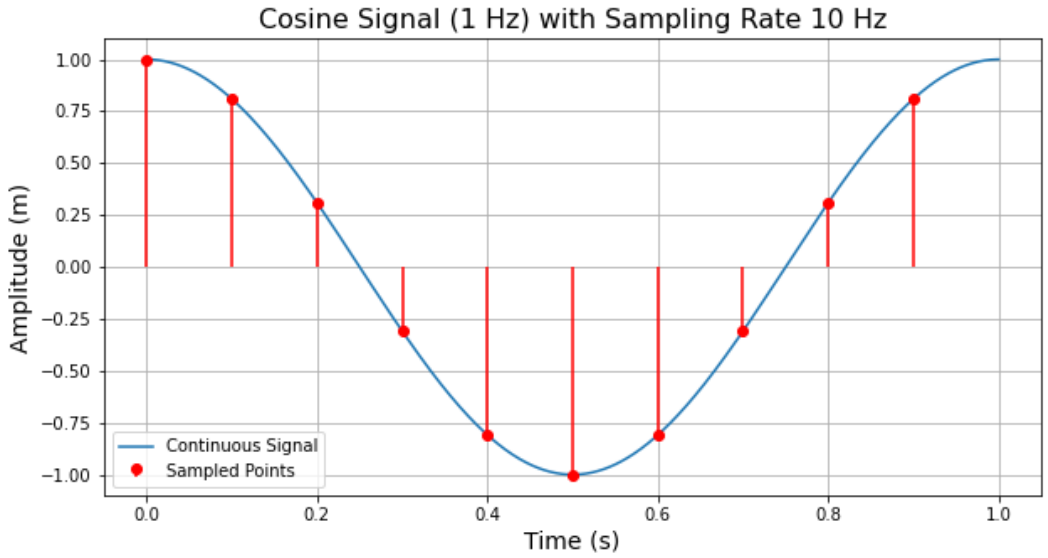
$$+ 0.809\left(\cos\left(\frac{2\pi \times 1 \times 1}{10}\right) - i \sin\left(\frac{2\pi \times 1 \times 1}{10}\right)\right)$$

$$+ 0.309\left(\cos\left(\frac{2\pi \times 1 \times 2}{10}\right) - i \sin\left(\frac{2\pi \times 1 \times 2}{10}\right)\right)$$

$$+ \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

$$+ 0.809\left(\cos\left(\frac{2\pi \times 1 \times 9}{10}\right) - i \sin\left(\frac{2\pi \times 1 \times 9}{10}\right)\right)$$

Example 1



Sample 0: Time = 0.00 s, Value = 1.000
Sample 1: Time = 0.10 s, Value = 0.809
Sample 2: Time = 0.20 s, Value = 0.309
Sample 3: Time = 0.30 s, Value = -0.309
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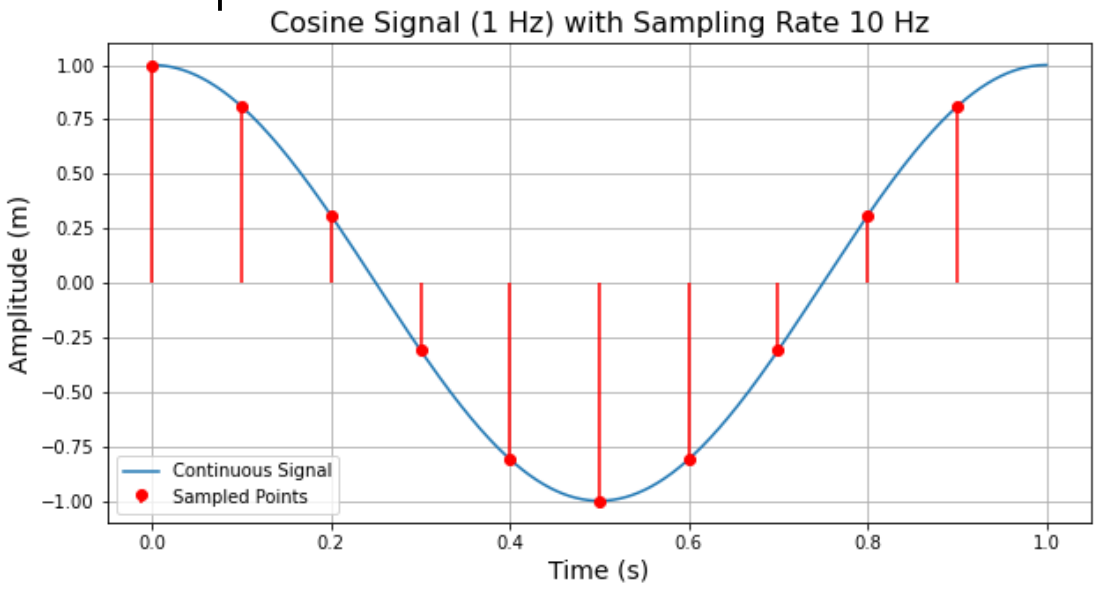
Sample 5: Time = 0.50 s, Value = -1.000
Sample 6: Time = 0.60 s, Value = -0.809
Sample 7: Time = 0.70 s, Value = -0.309
Sample 8: Time = 0.80 s, Value = 0.309
Sample 9: Time = 0.90 s, Value = 0.809

$$X_k = \sum_{n=0}^{N-1} x_n \left(\cos\left(\frac{2\pi kn}{N}\right) - i \sin\left(\frac{2\pi kn}{N}\right) \right)$$

$$\begin{aligned}
X_0 &= 0.000 + 0.000j \\
X_1 &= 5.000 + 0.000j \\
X_2 &= 0.000 + 0.000j \\
X_3 &= 0.000 + 0.000j \\
X_4 &= 0.000 + 0.000j \\
X_5 &= -0.000 + 0.000j \\
X_6 &= 0.000 + 0.000j \\
X_7 &= 0.000 + 0.000j \\
X_8 &= 0.000 + 0.000j \\
X_9 &= 5.000 + 0.000j
\end{aligned}$$

Example 1



$$
\begin{aligned}
X_0 &= 0.000 + 0.000j \rightarrow |X_0| = 0 \\
X_1 &= 5.000 + 0.000j \rightarrow |X_1| = 5 \\
X_2 &= 0.000 + 0.000j \rightarrow |X_2| = 0 \\
X_3 &= 0.000 + 0.000j \rightarrow |X_3| = 0 \\
X_4 &= 0.000 + 0.000j \rightarrow |X_4| = 0 \\
X_5 &= -0.000 + 0.000j \rightarrow |X_5| = 0 \\
X_6 &= 0.000 + 0.000j \rightarrow |X_6| = 0 \\
X_7 &= 0.000 + 0.000j \rightarrow |X_7| = 0 \\
X_8 &= 0.000 + 0.000j \rightarrow |X_8| = 0 \\
X_9 &= 5.000 + 0.000j \rightarrow |X_9| = 5
\end{aligned}
$$

Sample 0: Time = 0.00 s, Value = 1.000
Sample 1: Time = 0.10 s, Value = 0.809
Sample 2: Time = 0.20 s, Value = 0.309
Sample 3: Time = 0.30 s, Value = -0.309
Sample 4: Time = 0.40 s, Value = -0.809
Sample 5: Time = 0.50 s, Value = -1.000
Sample 6: Time = 0.60 s, Value = -0.809
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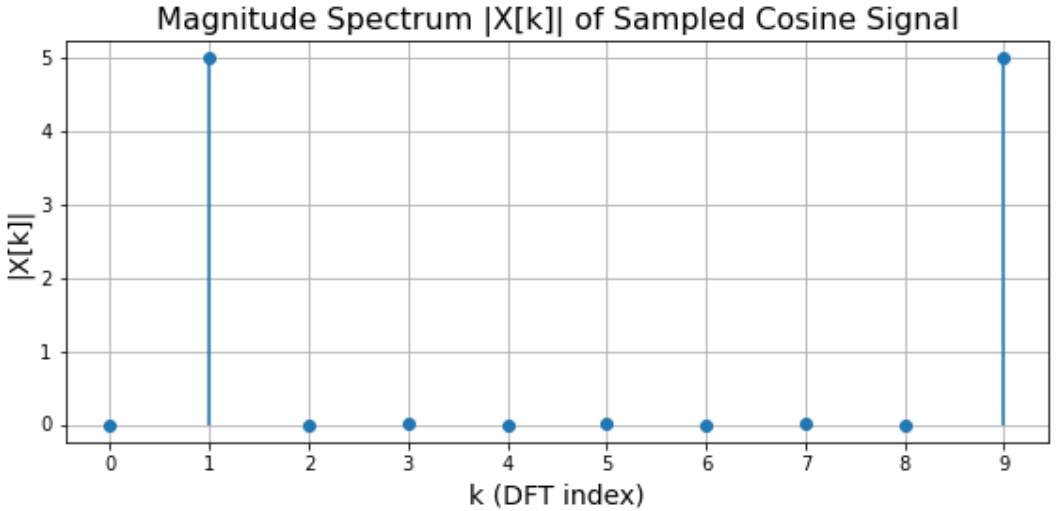
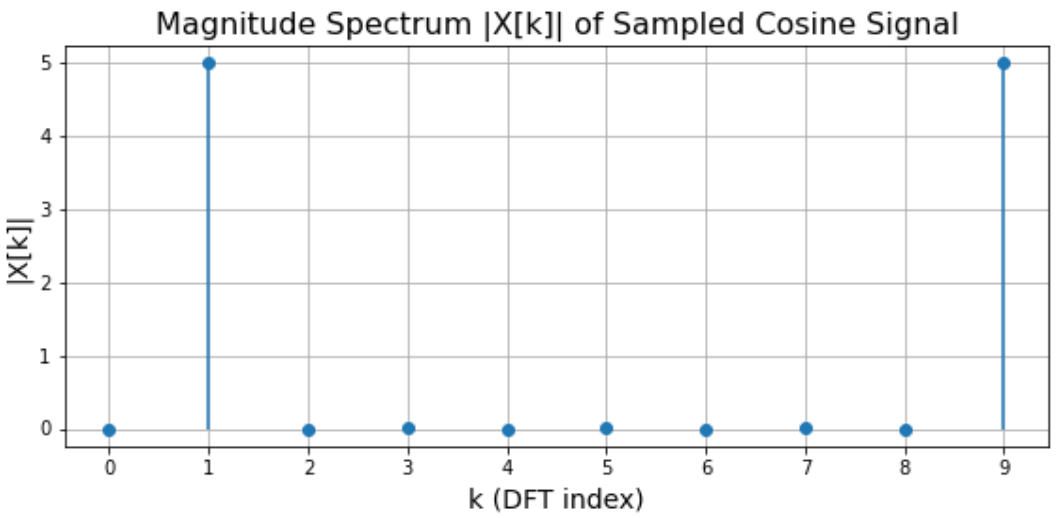
Sample 8: Time = 0.80 s, Value = 0.309
Sample 9: Time = 0.90 s, Value = 0.809



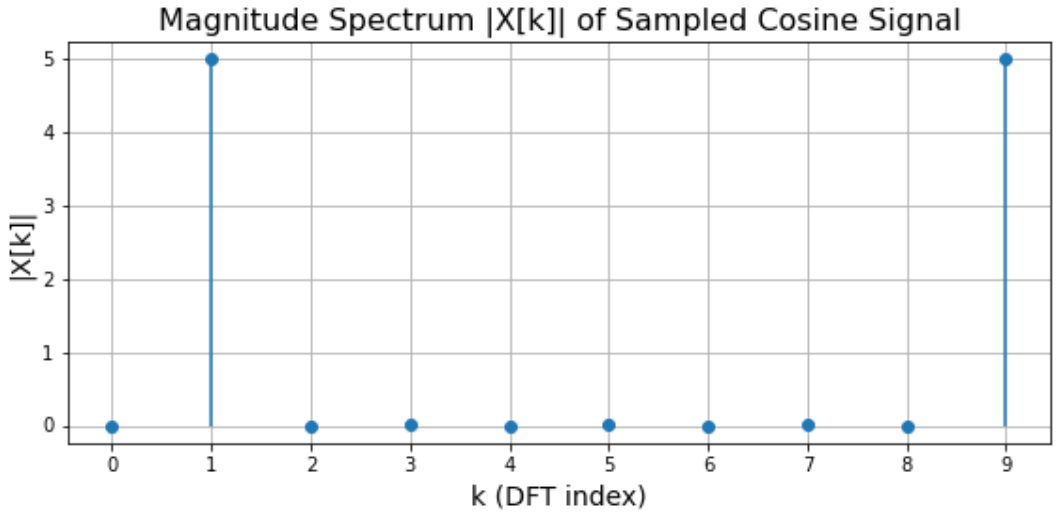
Example 1



When you compute a DFT with $N = 10$ samples, it breaks the spectrum into N frequency bins

- But for real signals, the spectrum is periodic and symmetric around **Nyquist frequency** $f_s/2 = 5$ Hz.
- Frequencies above $k = 5$(i.e., $k = 6$ to 9) are mirror images of $k = 1$ to 4.
- Therefore, a single side will be sufficient to analyze the frequency.

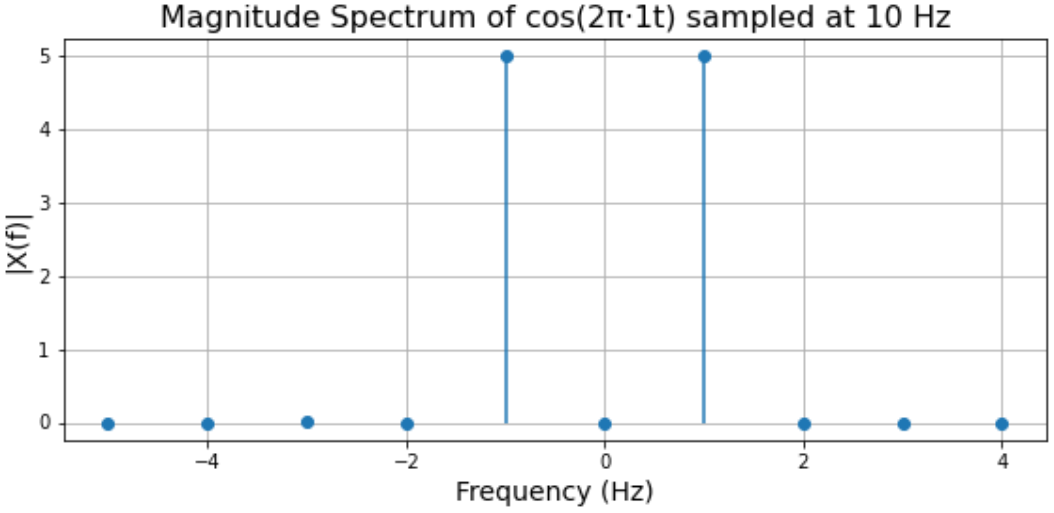
Nyquist frequency



This function returns the physical frequencies corresponding to each bin k, based on the sampling interval $d = 1/fs$. The formula is:

$$f_k = \begin{cases} \frac{k}{N} f_s & \text{if } k < \frac{N}{2} \\ \left(\frac{k-N}{N}\right) f_s & \text{if } k \geq \frac{N}{2} \end{cases}$$

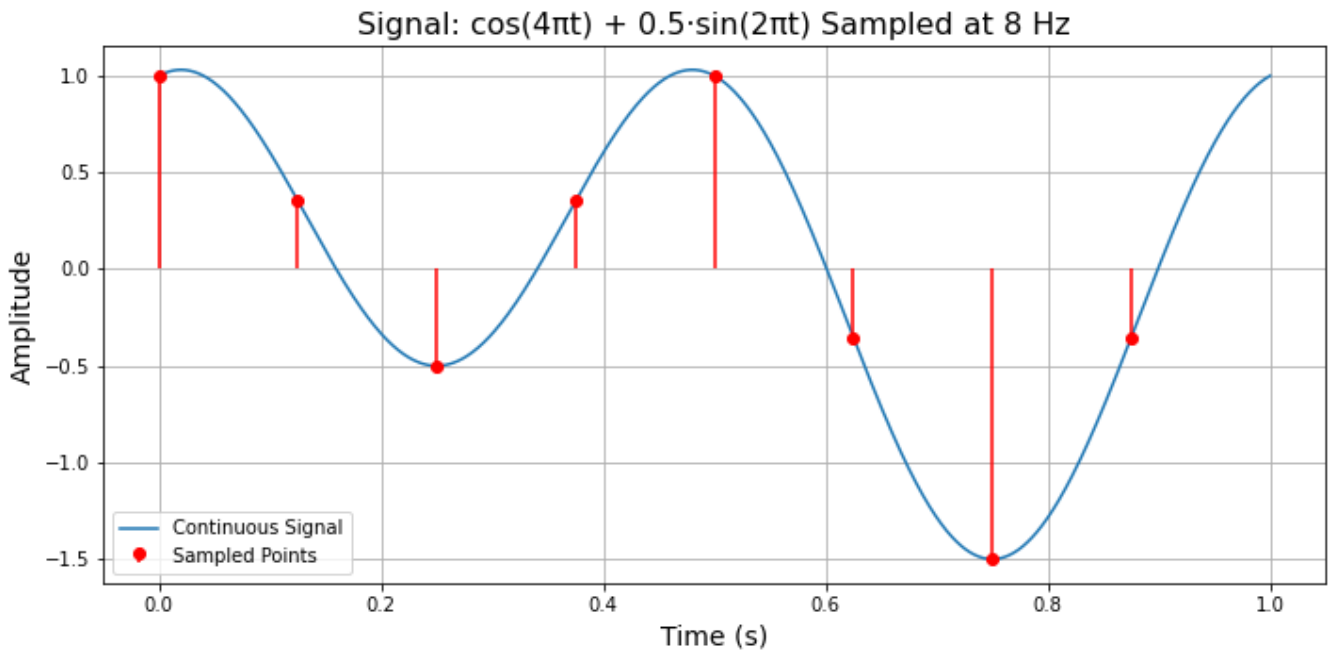
Nyquist frequency $= f_s/2$



The above graph is plotted against the frequency.

Negative frequencies were shifted to the right side of the graph.

Example 2

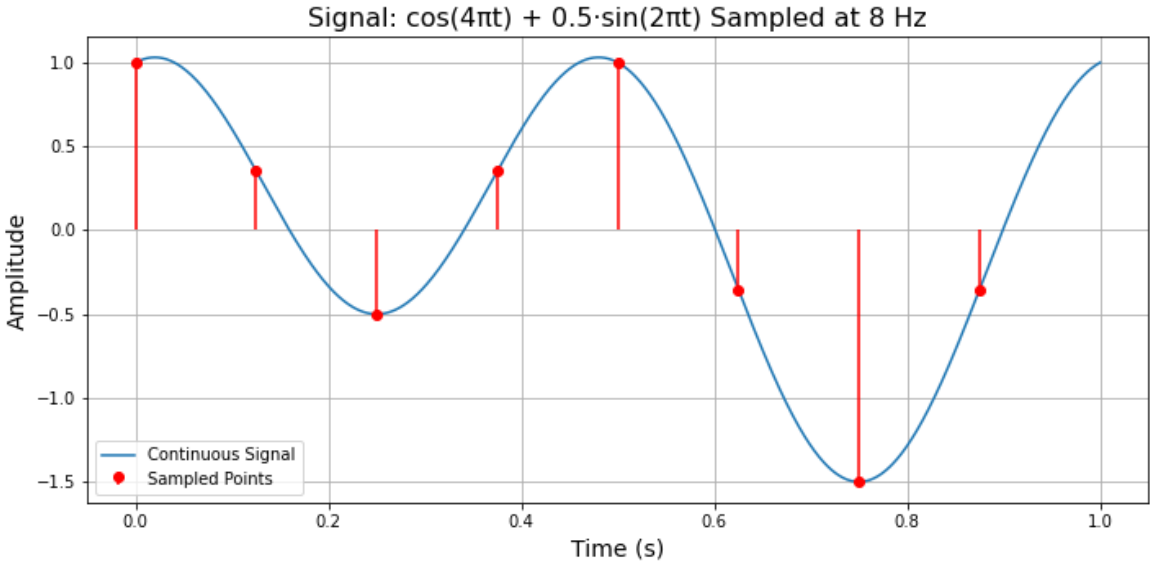


$$S = \mathrm{A}_1 \cos(2\pi f_1 t) + A_2 \sin(2\pi f_2 t)$$

$f_1 = 2\text{Hz}$ $f_2 = 1\text{Hz}$ $\mathrm{A}_1 = 1\text{mm}$ $\mathrm{A}_2 = 2\text{mm}$

Sampling Frequency $f_s = 8\text{Hz}$

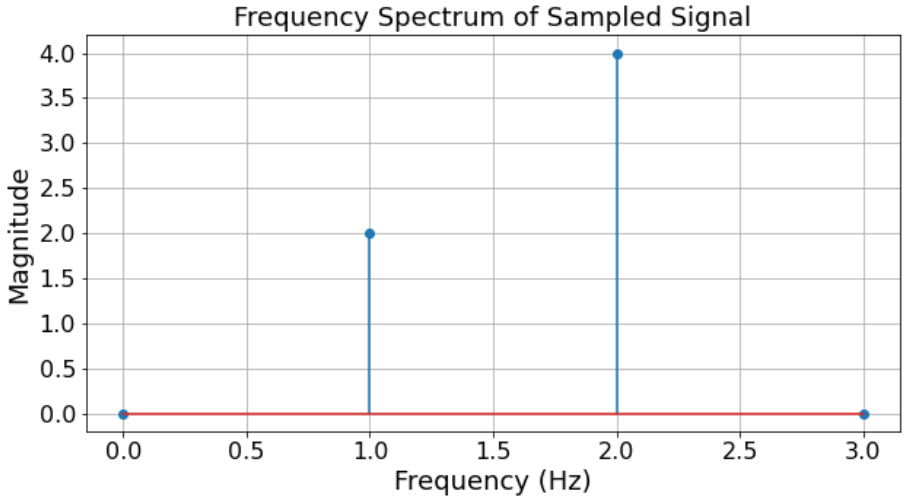
Example 2



Sample 0: t = 0.000 s, Value = 1.000
Sample 1: t = 0.125 s, Value = 0.354
Sample 2: t = 0.250 s, Value = -0.500
Sample 3: t = 0.375 s, Value = 0.354
Sample 4: t = 0.500 s, Value = 1.000
Sample 5: t = 0.625 s, Value = -0.354
Sample 6: t = 0.750 s, Value = -1.500
Sample 7: t = 0.875 s, Value = -0.354

$$X_k = \sum_{n=0}^{N-1} x_n \left(\cos\left(\frac{2\pi kn}{N}\right) - i \sin\left(\frac{2\pi kn}{N}\right) \right)$$

```
Bin     DFT Value (Complex)                 Amplitude
0       (-2.220446049250313e-16+0j)         0.0000
1       (-4.996003610813204e-16-2j)         2.0000
2       (4-1.1102230246251565e-15j)         4.0000
3       (4.996003610813204e-16-2.220446049250313e-16j)  0.0000
4       (2.220446049250313e-16+0j)          0.0000
5       (4.996003610813204e-16+2.220446049250313e-16j)  0.0000
6       (4+1.1102230246251565e-15j)         4.0000
7       (-4.996003610813204e-16+2j)         2.0000
```



$$f_k = \frac{k}{N} \cdot f_s = \frac{k}{8} \times 8 = k\text{Hz}$$